

Khwaja Yunus Ali University

Founder: Dr. M. M. Amjad Hussain

School of Science & Engineering

Department of CSE & EEE

Spring - 2025

Course Outline: Physics – II

Part – A:

1. **Course Code:** CSE 0533 1201 (for CSE) & PHY 0533 1201 (for EEE)
2. **Course Title:** Physics - II
3. **Course Type:** Related
4. **Level:** 1.2 (1st Year 2nd Semester)
5. **Academic Session:** Spring - 2025
6. **Course Instructor:** Md. Matiur Rahman
7. **Prerequisites:** N/A
8. **Credit Value:** 3.00
9. **Contact hours/week:** 3.00 hours
10. **Total Marks:** 100
11. **Rationale of the Course:** This course is to give students a broad understanding of physics and a foundation in problem solving and critical thinking. Topics covered in this course include Thermodynamics, Modern Physics and Optics.

12. Course Objectives

The principles of physics are the basis of all modern technology, and understanding the concepts of physics and knowing how to solve physics problems is a key indicator of success in advanced study in all technical fields, including biology, medicine, and the health sciences. The purpose this course is to give students a broad understanding of physics and a foundation in problem solving and critical thinking. Topics covered in this course include Thermodynamics, Modern Physics and Optics.

13. Course Learning Outcomes (CLOs) and its mapping with PLOs:

CLOs	After successful completing this course, students will be able to.	PLO Mapping
CLO1	Specify the basic terms related to thermodynamics, modern physics and light.	PLO 1
CLO2	Identify and formulate the various quantities to explain physical and engineering phenomena.	PLO 1
CLO3	Apply critical thinking and problem-solving skills to solve a variety of problems concerning thermodynamics, modern physics and light.	PLO 2
CLO4	Demonstrate analytical proficiency to establish the physical and mathematical logic behind natural phenomena observed in everyday life.	PLO 4

Part – B:

14. Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs:

Week	Topic	Teaching Strategy	Assessment Technique	Alignment with CLOs
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01	Principle of temperature measurements: Platinum resistance thermometer, Thermo-electric thermometer, Pyrometer	Interactive discussion Demonstration Group activity Tutorials	Oral Test Class works Assignment Mid-term Exam	1 & 2
02	Kinetic theory of gases: Maxwell's distribution of molecular speeds, Mean free path, Equipartition of energy, Brownian motion, van der Waal's equation of state	Interactive discussion Group activity Tutorials Problem Solving	Oral Test Home works Assignment Mid-term Exam	2 & 3
03	Review of the First law of thermodynamics and its application	Interactive discussion Group Presentation Tutorials	Class Test Home works Mid-term Exam	1
04	Reversible & irreversible processes, Second law of thermodynamics, Carnot; Efficiency of heat engines, Carnot theorem, Entropy and Disorder	Interactive discussion Demonstration Group activity Tutorials	Oral Test Class works Mid-term Exam	2
05	Thermodynamic Functions, Maxwell relations, Clausius-Clapeyron equation, Gibbs phase rule, Third law of thermodynamics	Interactive discussion Demonstration Group activity Tutorials	Oral Test Home works Assignment Mid-term Exam	3 & 4
06	Interpretation of Bohr's postulates, Properties of nucleus, nuclear reactions, Fission. Fusion, Chain reaction, nuclear reactor	Interactive discussion Demonstration Group activity Tutorials	Oral Test Class works Mid-term Exam	1
07	Wave particle duality, Compton effect, Photo-electric effect, Quantum theory of radiation	Interactive discussion Demonstration Group activity Tutorials	Class Test Home works Mid-term Exam	2
08	Radioactive disintegration	Interactive discussion Demonstration Group activity Tutorials	Oral Test Class works Assignment Final Exam	1 & 3
09	Michelson-Morley's experiment, Frame of references, Galilean transformation, Lorentz transformations	Interactive discussion Group activity Tutorials Problem Solving	Oral Test Home works Assignment Final Exam	3 & 4
10	Special theory of relativity & its consequences	Interactive discussion Group Presentation Tutorials	Class Test Home works Final Exam	3 & 4
11	Interference of light, Young's double slit experiment, Displacements of fringes & its uses. Newton's rings, Thin film	Interactive discussion Demonstration Group activity Tutorials	Oral Test Class works Final Exam	1 & 2
12	Diffraction of light: Fresnel and Fraunhofer diffraction. Diffraction by single slit, Resolving power, Diffraction grating.	Interactive discussion Demonstration Group activity Tutorials	Oral Test Home works Assignment Final Exam	2 & 3
13	Polarization: Production & analysis of polarized light, Brewster's law, Malus law, Polarization by double refraction.	Interactive discussion Group activity Tutorials Problem Solving	Oral Test Class works Final Exam	2

14	Nicol prism, Mathematical Problems	Interactive discussion Group Presentation Tutorials	Class Test Home works Final Exam	3 & 4
15	Review Class/Problem Solving	Interactive discussion Group Presentation Tutorials	Oral Test Class works Final Exam	2 & 4

Part – C:

15. Assessment and Evaluation

1. Assessment and Evaluation

1. Assessment Strategy

Students are evaluated out of total 100 Marks. There are different types of assessment tools. In the **Mid Term** examination, total time is **1.5 hours** where the marks is **30**. In the **Final Examination**, total marks is **40** and allotted time is **2 Hours**. The detail with **CLOs-Assessment Mapping** is given below.

CLOs	Continuous Assessment (30)			Summative Assessment (70)				Total (100)
	Class Performance (10)	Class Test (10)	Assignment (10)	Mid-term Exam (30)		Final Exam (40)		
				MCQ (10)	SAQ (20)	MCQ (10)	SAQ (30)	
CLO-1	4	-	-	4	5	4	-	17
CLO-2	3	5	-	3	5	3	10	29
CLO-3	3	5	5	2	10	2	10	37
CLO-4	-	-	5	1	-	1	10	17

2. Evaluation System (Grading System)

Numerical Number	Letter Grade		Grade Point
80 % and above	A+	(A Plus)	4.00
75 % to less than 80 %	A	(A Regular)	3.75
70 % to less than 75 %	A-	(A Minus)	3.50
65 % to less than 70 %	B+	(B Plus)	3.25
60 % to less than 65 %	B	(B Regular)	3.00
55 % to less than 60 %	B-	(B Minus)	2.75
50 % to less than 55 %	C+	(C Plus)	2.50
45 % to less than 50 %	C	(C Regular)	2.25
40 % to less than 45 %	D		2.00
Less than 40 %	F		0.00

3. Make up procedures:

For weak students, we take extra class before their midterm and final examinations. Beside this, we counseling with them and solve their problem as soon as possible.

Part – D:

16. Learning Materials:

1) Recommended Reading

- Physics for Engineers (Part – 1 & Part – 2) – Gias Uddin Ahmed

- Halliday Resnick - Fundamental of Physics, 2nd edition
- Brijlal - Heat and Thermodynamics - 1st edition
- Tofazzal Hossain - A Text Book of Heat, 2nd edition
- Beiser - Concept of Modern Physics, 6th edition

2) Supplementary Reading

- Web Source Link
- Journal papers
- Physics for Engineers – Jiaul Ahsan

3) Others:

- Lecture Outlines
- PowerPoint File
- Webpage